



FINAL YEAR PROJECT PROPOSAL

Project I (3.0 Credits) & Project II (3.0 Credits)

Design and Implementation of a Secure Enterprise Infrastructure (ApexCorp Ltd.) Using Cisco Network Architecture, Windows Server 2022, and Sophos XG Next-Generation Firewall

Submitted By

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Executive Summary & Background

Modern enterprises are increasingly dependent on secure, scalable, and centrally managed IT infrastructures. However, many organizations continue to operate flat and unsegmented networks that expose critical business resources to cyber threats, unauthorized access, and operational disruptions.

ApexCorp Ltd. is a hypothetical mid-sized enterprise designed for this academic final year project to simulate real-world corporate infrastructure requirements. The organization requires a secure and scalable network architecture that supports centralized user authentication, network segmentation, enterprise firewall policies, and high availability services.

The proposed infrastructure combines industry-standard technologies including Cisco enterprise networking concepts, Windows Server 2022 services, and Sophos XG Next-Generation Firewall capabilities to develop a practical enterprise security architecture.

The project will implement both simulated and live enterprise environments through Cisco Packet Tracer and virtualization technologies such as VirtualBox or VMware. The infrastructure will incorporate:

- Centralized Identity Management.
- Active Directory-based user authentication.
- VLAN-based network segmentation.
- Sophos Next-Generation Firewall security policies.
- DNS and DHCP services.
- Group Policy-based endpoint hardening.
- Identity-aware Layer-7 security policies.
- Enterprise-grade ransomware and malicious traffic mitigation mechanisms.

This project aims to provide practical exposure to modern enterprise infrastructure deployment and cybersecurity implementation practices widely adopted in corporate environments.

Problem Statement

Many organizations continue to face significant cybersecurity challenges due to:

- Flat network architectures.
- Lack of centralized identity management.
- Weak perimeter security.
- Absence of access control mechanisms.
- Improper segmentation of users and servers.

- Inadequate endpoint security policies.
- Limited visibility into network traffic.

Without enterprise-grade security controls, organizations remain vulnerable to:

- Unauthorized access.
- Insider threats.
- Malware and ransomware attacks.
- Lateral movement attacks.
- Credential misuse.
- Service disruptions.

The proposed solution addresses these challenges through the design and

implementation of a secure enterprise infrastructure utilizing Cisco networking technologies, Microsoft Windows Server

services, and Sophos Next-Generation Firewall capabilities.

Project Objectives

The primary objective of this project is to design and implement a secure and centrally managed enterprise network infrastructure.

General Objectives

- Design an enterprise-grade network architecture.
- Implement secure network segmentation using VLANs.
- Deploy centralized identity management using Active Directory.
- Configure enterprise DNS and DHCP services.
- Implement perimeter security using Sophos XG Firewall.
- Develop role-based access control policies.
- Demonstrate enterprise cybersecurity best practices.

Specific Objectives

Centralized Identity Management

- Deploy Active Directory Domain Services.
- Configure user and computer authentication.
- Implement Organizational Units and security groups.

Network Segmentation

- Design VLAN-based network architecture.

- Configure Inter-VLAN routing.
- Implement STP and HSRP concepts.
- Separate client, server, and management networks.

Perimeter Security

- Configure Sophos Next-Generation Firewall.
- Establish WAN, LAN, and DMZ zones.
- Configure IPS and application control.
- Implement secure Internet access policies.

Identity-Aware Security Policies

- Integrate Sophos Firewall with Active Directory.
- Configure user-based firewall rules.
- Apply role-based web filtering policies.

Group Policy Hardening

- Configure Group Policy Objects (GPOs).
- Apply password and account lockout policies.
- Implement Local Administrator Password Solution (LAPS).
- Harden domain-joined client systems.

Dual-Phase Implementation Plan

The final year project is divided into two academic phases.

PROJECT I (3.0 Credits)

Phase Title

Enterprise Network Design and Core Infrastructure Deployment

Scope of Work

A. Cisco Network Topology Design

The network architecture will be designed using Cisco Packet Tracer and will include:

- VLAN Segmentation
- Inter-VLAN Routing
- Spanning Tree Protocol (STP)
- Hot Standby Router Protocol (HSRP)
- Enterprise Switch Configuration

B. Windows Server 2022 Deployment

The following services will be deployed:

- Active Directory Domain Services (AD DS)
- Domain Controller Configuration
- DNS Server
- DHCP Server
- Organizational Units
- User and Group Management

C. Sophos Firewall Baseline Configuration

The firewall will be configured with:

- WAN Interface
- LAN Interface
- IP addressing plan.
- Initial project presentation.

- Initial Firewall Rules
- NAT Policies
- Internet Connectivity
- Network Address Translation

D. Domain Workstation Deployment

Client systems will include:

- Windows 10 Pro
- Windows 10 Pro N
- Domain Joining Procedures
- User Authentication Testing

Deliverables for Project I

- Cisco Packet Tracer (.pkt) topology file.
- Enterprise network design documentation.
- Windows Server deployment documentation.
- Sophos Firewall baseline configuration.
- Domain-joined client implementation.

PROJECT II (3.0 Credits)

Phase Title

Enterprise Security Integration and Cybersecurity Hardening

Scope of Work

A. Sophos and Active Directory Integration

The firewall will be integrated with:

- Active Directory
- LDAP Authentication
- Sophos Transparent Authentication Suite (STAS)
- Identity-aware policies

B. Layer-7 Security Policies

Implementation of:

- Web Filtering
- Application Filtering
- User-based Internet policies
- Department-based access control

C. Group Policy Security Hardening

The Windows domain environment will include:

- Password Policies
- Account Lockout Policies
- Windows Security Policies
- USB Restrictions
- Firewall Policies
- Login Restrictions

D. Microsoft LAPS Implementation

Implementation objectives include:

- Automatic local administrator password management.
- Password rotation.
- Secure administrative access control.

E. Threat Detection and Demonstration

Practical demonstrations will include:

- Malicious website blocking.
- Unauthorized application blocking.

- Policy enforcement.
- Simulated ransomware or malware prevention scenarios.
- Security event monitoring.

Deliverables for Project II

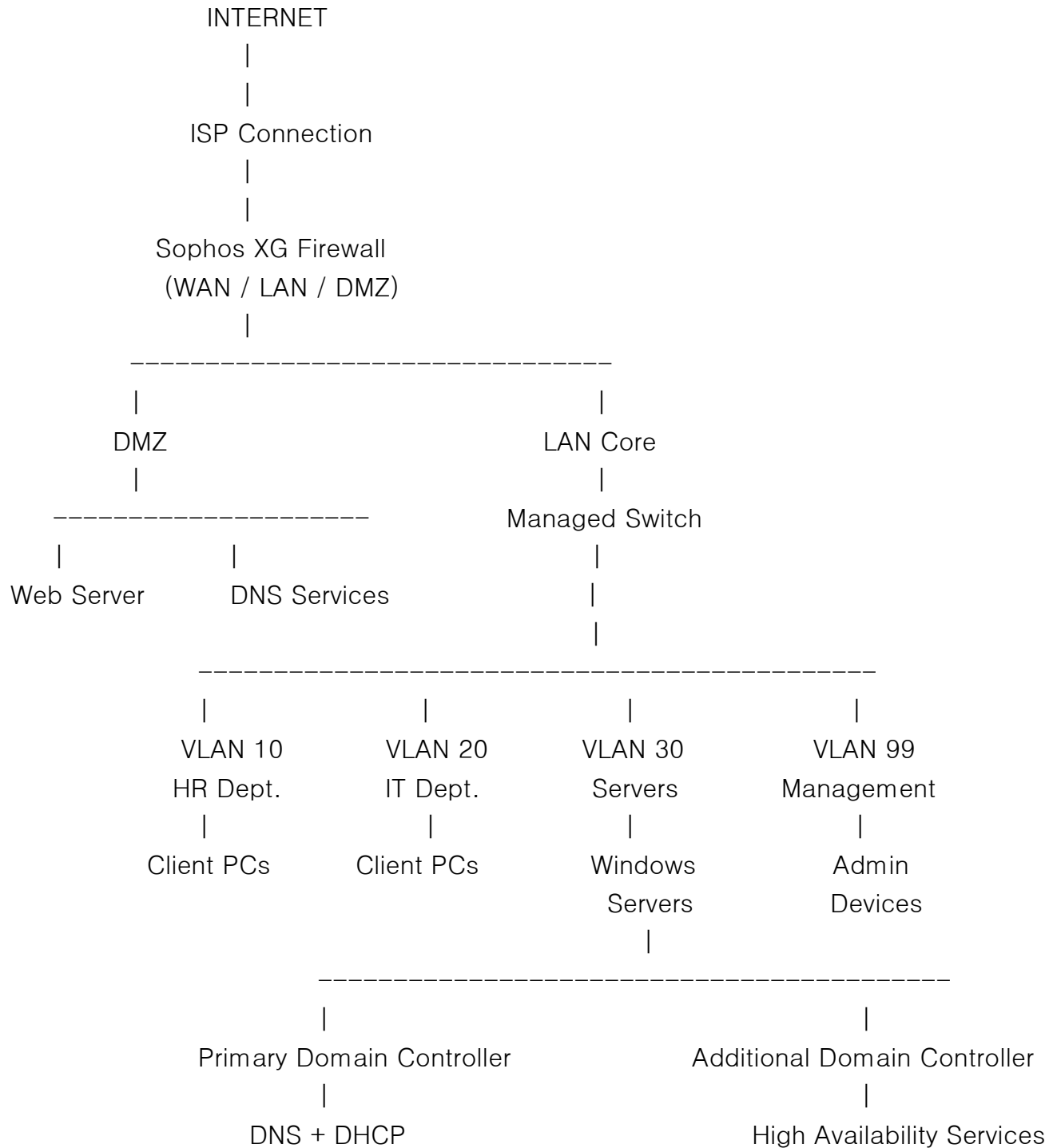
- Sophos-AD integration documentation.
- Security policy configuration reports.
- Group Policy configuration documentation.
- Firewall screenshots and implementation evidence.
- Threat demonstration report.
- Final project presentation.
- Complete technical documentation.

Software Specifications

Software	Version
Windows Server	Windows Server 2022 Standard
Windows Client OS	Windows 10 Pro/ Pro N
Sophos Firewall	Sophos Firewall OS (SFOS) v21 or Latest Stable Release
Cisco Packet Tracer	Version 8.x or Latest Stable Release
Virtualization Platform	Oracle VirtualBox 7.x
Alternative Platform	VMware Workstation Pro
Documentation Tools	Microsoft Word & Visio
Network Simulation	Cisco Packet Tracer

System Architecture & Network Design

Enterprise Infrastructure Architecture



Expected Deliverables & Business Value

Academic Deliverables

The final project submission will include:

- Cisco Packet Tracer topology file.
- Network architecture diagrams.

- Windows Server implementation.
- Sophos Firewall configurations.
- Domain infrastructure implementation.
- VLAN and subnetting plans.
- Security policy documentation.
- Threat demonstration report.
- Project presentation slides.
- Final technical report.
- Centralized authentication and authorization.
- Reduced attack surface through VLAN segmentation.
- Identity-aware firewall policies.
- Improved endpoint security.
- Enhanced network visibility and monitoring.

Practical Learning Outcomes

Upon successful completion, the project will demonstrate:

- Enterprise network design principles.
- Active Directory deployment.
- Network segmentation techniques.
- Firewall policy management.
- Identity-aware access control.
- Security hardening practices.
- Cybersecurity incident prevention mechanisms.
- Infrastructure documentation standards.

Business Value

The proposed enterprise infrastructure provides the following advantages:

Security Benefits

Operational Benefits

- Simplified infrastructure management.
- Improved scalability.
- High availability support.
- Policy-driven access control.
- Efficient user and device management.

Educational and Industry Relevance

This final year project reflects modern enterprise infrastructure and cybersecurity practices used in corporate environments. The implementation aligns with industry requirements for Network Security Engineers, System Administrators, SOC Analysts, and Infrastructure Security Professionals. It provides practical, hands-on experience in designing and securing enterprise environments using widely adopted technologies.

Conclusion

The "Design and Implementation of a Secure Enterprise Infrastructure (ApexCorp Ltd.) Using Cisco Network Architecture, Windows Server 2022, and Sophos XG Next-Generation Firewall" project aims to bridge academic knowledge with real-world enterprise infrastructure practices. By integrating network architecture, centralized identity management, next-generation firewall technologies, and cybersecurity hardening techniques, the project delivers a comprehensive and scalable enterprise security solution suitable for both academic evaluation and professional skill development. The successful implementation of this project will demonstrate competencies in enterprise networking, system administration, and cybersecurity, while providing practical experience aligned with current industry standards and organizational security requirements.